

Robustel MQTT Device Setup & Commissioning Procedure

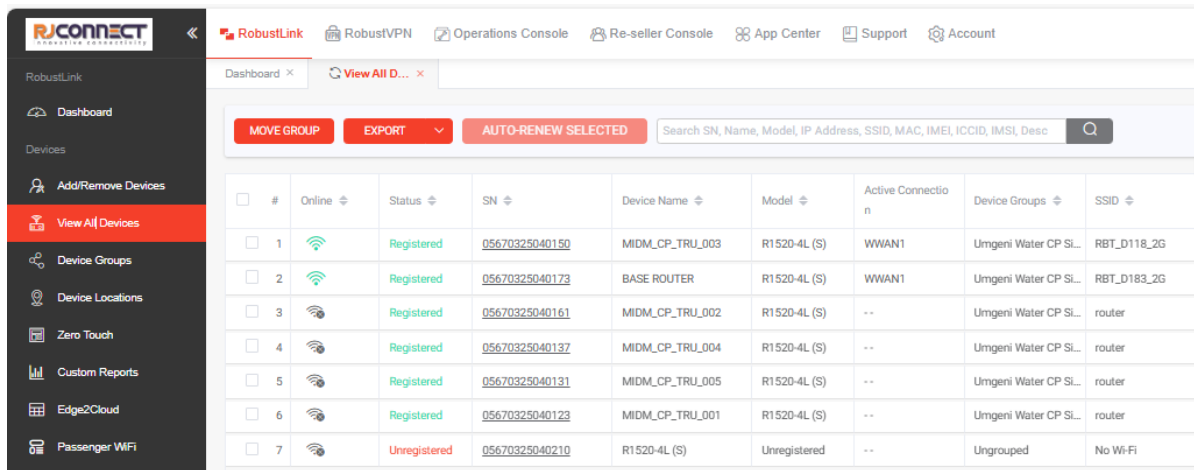
Drafted By	Date	Rev
MC Vermaak	2025/07/21	1

* All required device access credentials and supporting apps can be found here:
 \\Adtech-dfs\dfs\Adroit Support\Adroit Projects\02 - Smart UI\Umgeni\CP MQTT to Fx Pilot

1. Device Online Status

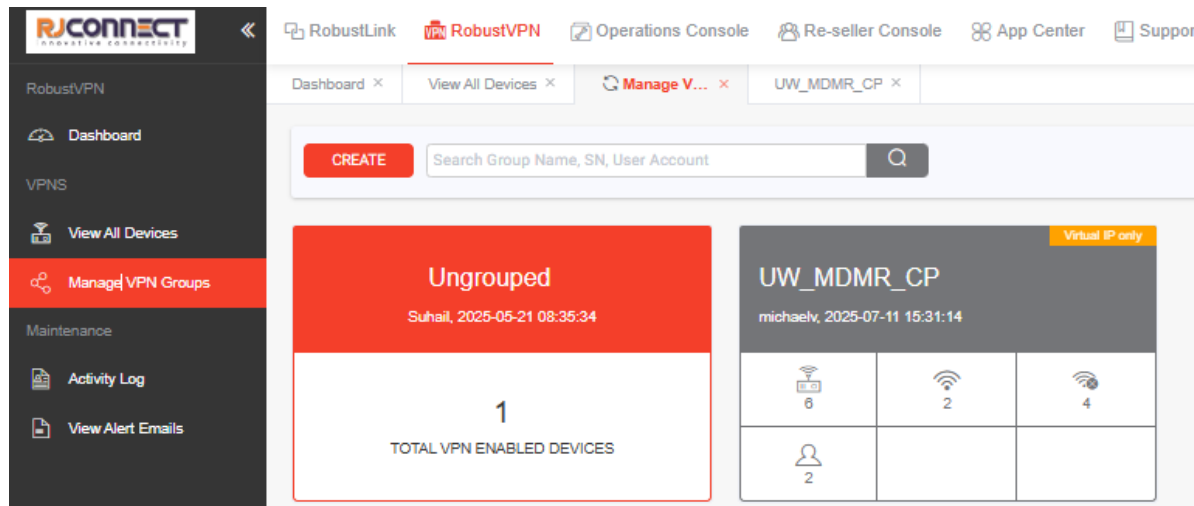
- 1.1. Login to Robustel RCMS Platform. Go to RobustelLink and view all devices. Verify that the router to be tested is online and registered to the network.

Follow link: [Robustel Cloud Manager Service](#)



	#	Online	Status	SN	Device Name	Model	Active Connection	Device Groups	SSID
☐	1		Registered	05670325040150	MIDM_CP_TRU_003	R1520-4L (S)	WWAN1	Umgeni Water CP SL...	RBT_D118_2G
☐	2		Registered	05670325040173	BASE ROUTER	R1520-4L (S)	WWAN1	Umgeni Water CP SL...	RBT_D183_2G
☐	3		Registered	05670325040161	MIDM_CP_TRU_002	R1520-4L (S)	--	Umgeni Water CP SL...	router
☐	4		Registered	05670325040137	MIDM_CP_TRU_004	R1520-4L (S)	--	Umgeni Water CP SL...	router
☐	5		Registered	05670325040131	MIDM_CP_TRU_005	R1520-4L (S)	--	Umgeni Water CP SL...	router
☐	6		Registered	05670325040123	MIDM_CP_TRU_001	R1520-4L (S)	--	Umgeni Water CP SL...	router
☐	7		Unregistered	05670325040210	R1520-4L (S)	Unregistered	--	Ungrouped	No Wi-Fi

- 1.2. Go to RobustelVPN and select UW_MDMR_CP VPN group.



- 1.3. Click on Devices. See that the router to be tested is on the network, allocated to the same VPN group and connected to the VPN network for remote management.

TOTAL VPN ENABLED DEVICES

6

VPNS CONNECTED

2

VPNS DISCONNECTED

4

USERS

2

Information

Devices

Users

Settings

Alerts

Connectivity Logs

End Device Management

ADD

REMOVE

INITIALIZE

DATA LIMIT

Search SN, Name, IP Added By

Q

1/1

←

→

#	VPN Status	SN	Device Name	Device OS	Virtual IP	Total VPN Data	Last Connected Date	Initialization Status	Monthly VPN Data	VPN Data Limit	License	Tags	Connect VPN
1		05670325040150	MDM_CP_THU_003	RobustOS	10.245.164.5	9.420 MB	2025-07-21 15:05:30	Success	57.53MB	10GB	Advanced		
2		05670325040173	BASE ROUTER	RobustOS	10.245.164.8	21.203 MB	2025-07-21 15:15:20	Success	21.92MB	10GB	Advanced		
3		05670325040123	MDM_CP_THU_001	RobustOS	10.245.164.4	0 B		Success	30.29MB	10GB	Advanced		
4		05670325040131	MDM_CP_THU_005	RobustOS	10.245.164.9	0 B		Success	0B	10GB	Advanced		
5		05670325040137	MDM_CP_THU_004	RobustOS	10.245.164.7	0 B		Success	32.07KB	10GB	Advanced		
6		05670325040161	MDM_CP_THU_002	RobustOS	10.245.164.6	15.049 KB	2025-07-16 23:05:00	Success	1.12MB	10GB	Captive Portal PRO		

2. Router Remote Access

- 2.1. Go to RobustLink and view all devices. See that the router to be tested is online and registered to the network.
- 2.2. Click on the router's underlined serial number

RJCONNECT **RobustLink** RobustVPN Operations Console Re-seller Console App Center Support Account

Dashboard x View All D... x

MOVE GROUP EXPORT AUTO-RENEW SELECTED Search SN, Name, Model, IP Address, SSID, MAC, IMEI, ICCID, IMSI, Desc

	#	Online	Status	SN	Device Name	Model	Active Connection	Device Groups	SSID
☐	1		Registered	05670325040150	MIDM_CP_TRU_003	R1520-4L (S)	WWAN1	Umgeni Water CP SL...	RBT_D118_2G
☐	2		Registered	05670325040173	BASE ROUTER	R1520-4L (S)	WWAN1	Umgeni Water CP SL...	RBT_D183_2G
☐	3		Registered	05670325040161	MIDM_CP_TRU_002	R1520-4L (S)	--	Umgeni Water CP SL...	router
☐	4		Registered	05670325040137	MIDM_CP_TRU_004	R1520-4L (S)	--	Umgeni Water CP SL...	router
☐	5		Registered	05670325040131	MIDM_CP_TRU_005	R1520-4L (S)	--	Umgeni Water CP SL...	router
☐	6		Registered	05670325040123	MIDM_CP_TRU_001	R1520-4L (S)	--	Umgeni Water CP SL...	router
☐	7		Unregistered	05670325040210	R1520-4L (S)	Unregistered	--	Ungrouped	No Wi-Fi

2.3. Click on Remote Access

RJCONNECT **RobustLink** RobustVPN Operations Console Re-seller Console App Center Support Account

Dashboard x View All Devices x 05670325... x

MIDM_CP_TRU_003 ONLINE WWAN1 CONNECTION TYPE

Information Reports Firmware Apps Configuration Zero Touch Remote Actions Remote Access CLI Access Syslog Alert Logs Captive Portal

Status: Registered SN: 05670325040150

2.4. Generate an Access Link

RJCONNECT **RobustLink** RobustVPN Operations Console Re-seller Console App Center Support Account

Dashboard x View All Devices x 05670325... x

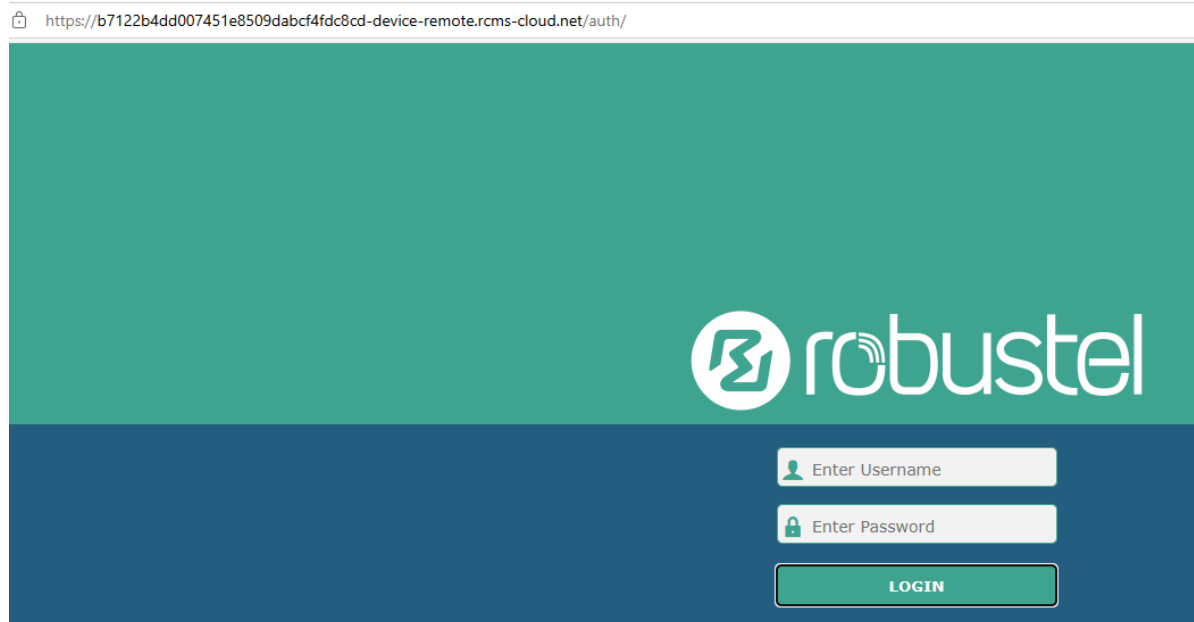
MIDM_CP_TRU_003 ONLINE WWAN1 CONNECTION TYPE 868.189 MB SIM1 0 B SIM2

Information Reports Firmware Apps Configuration Zero Touch Remote Actions Remote Access CLI Access Syslog Alert Logs Captive Portal

GENERATE LINK

#	Access Link	Protocol	Link Status	Generated Time	Validity Period
1	h7122b-d4d007451e8509dabc4f46d0d-device-remote-oms-cloud.net	HTTPS	Valid	2025-07-21 12:41:28	2025-07-23 12:41:27

- 2.5. Click on the generated link. This will launch a new webpage instance in your internet browser. Use the provided login credentials to access the router.



- 2.6. Click on status.
- Ensure that firmware 5.3.5 or higher is installed on the router. If not, then the router first needs to be updated.
 - Copy the device serial number to notepad. Will be used during the MQTT client setup.



[Save & Apply](#)
[Reboot](#)
[Logout](#)

Username : admin

Status

Interface

Network

VPN

Services

System

Edge2Cloud

Status

System Information

Internet Status

Modem Status

System Information

Device Model

R1520-4L (S)

System Uptime

0 days, 03:59:09

System Time

Mon Jul 21 16:33:25 2025

RAM Usage

54M Free/128M Total

Firmware Version

5.3.5 (f16daf0)

Cybersecurity Version

ROSCSBLV2

Hardware Version

1.2

Kernel Version

4.9.152

Serial Number

05670325040150

Internet Status

Uptime

0 days, 03:58:17

Active IPv4 Link

WWAN1

IPv4 Address

100.124.120.48/255.255.255.224

IPv4 Gateway

100.124.120.49

IPv4 DNS

41.1.111.253 41.1.114.13

Active IPv6 Link

Disabled

IPv6 Address

IPv6 Gateway

IPv6 DNS

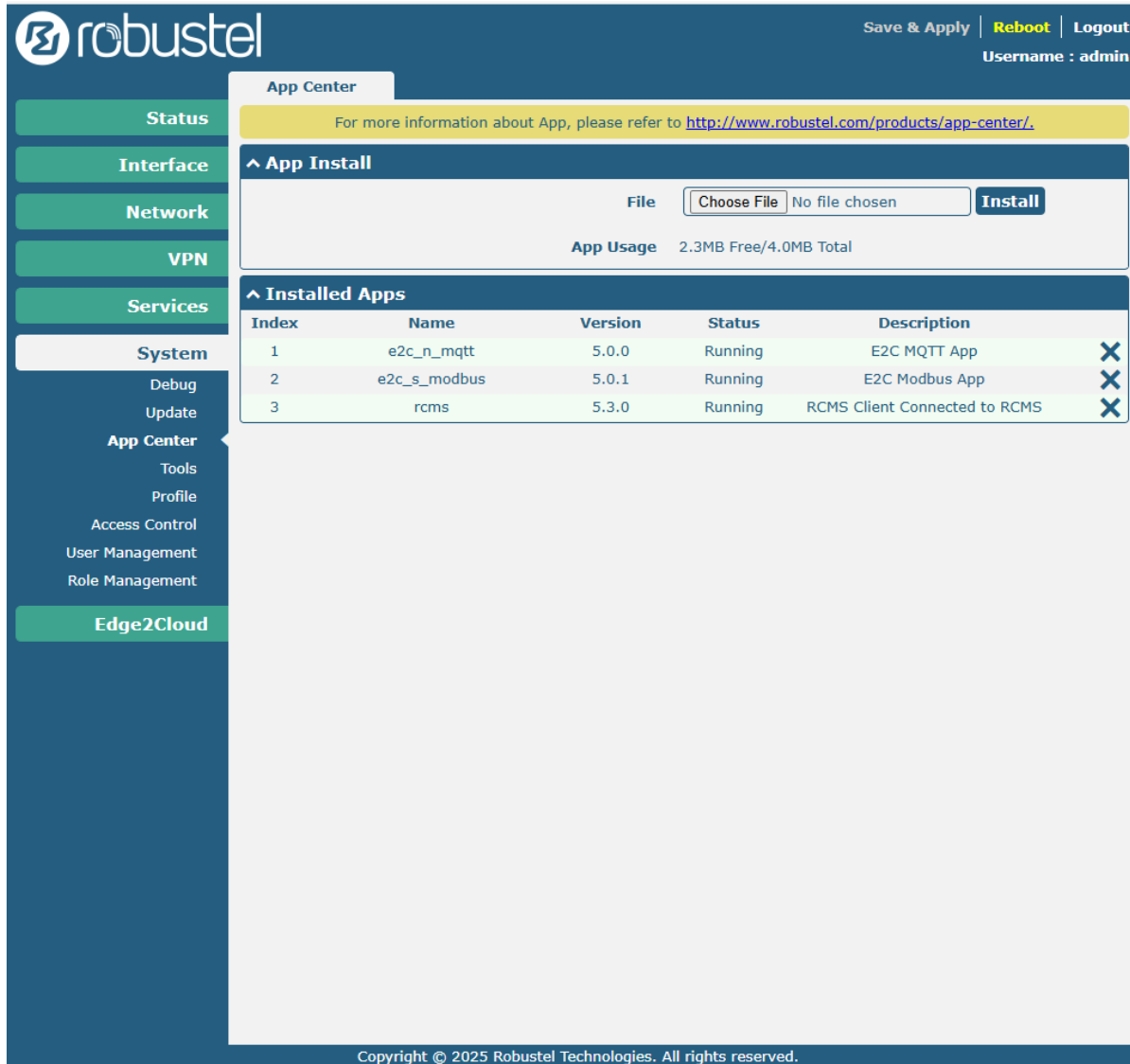
Modem Status

Modem Model

EC25-EC

Copyright © 2025 Robustel Technologies. All rights reserved.

- 2.7. Click on System > App Centre and verify the installed application & versions. If these app and equivalent versions or higher is not installed, then these apps first need to be installed or updated.

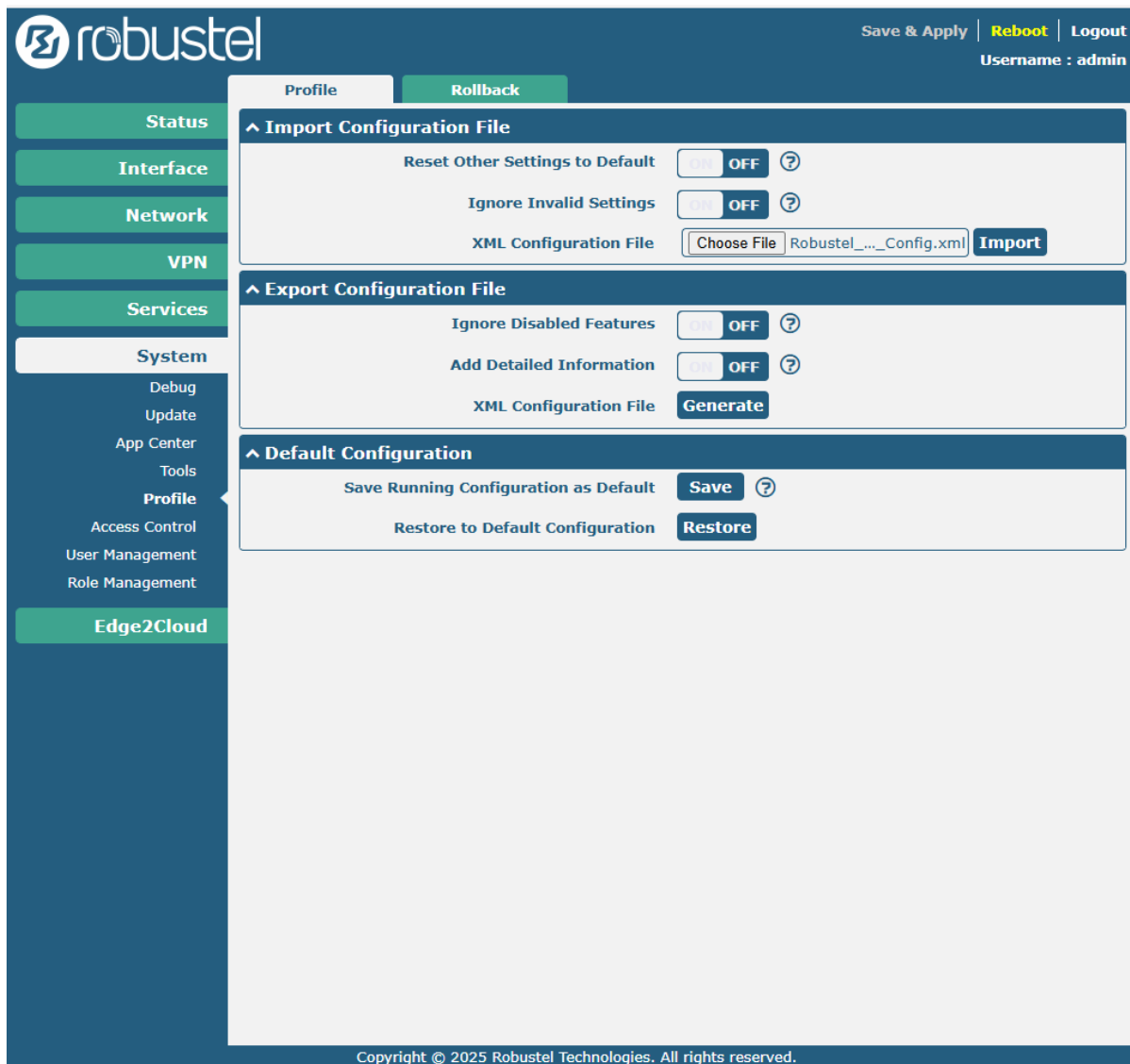


The screenshot shows the Robustel App Center interface. The top navigation bar includes 'Save & Apply', 'Reboot', and 'Logout' buttons, along with the 'Username : admin' display. The left sidebar contains a menu with 'Status', 'Interface', 'Network', 'VPN', 'Services', 'System', and 'Edge2Cloud'. The 'App Center' tab is active, displaying a yellow banner with a link to the Robustel App Center website. Below the banner, the 'App Install' section features a 'File' input field with a 'Choose File' button and an 'Install' button. The 'App Usage' section shows '2.3MB Free/4.0MB Total'. The 'Installed Apps' section contains a table with the following data:

Index	Name	Version	Status	Description	
1	e2c_n_mqtt	5.0.0	Running	E2C MQTT App	✕
2	e2c_s_modbus	5.0.1	Running	E2C Modbus App	✕
3	rcms	5.3.0	Running	RCMS Client Connected to RCMS	✕

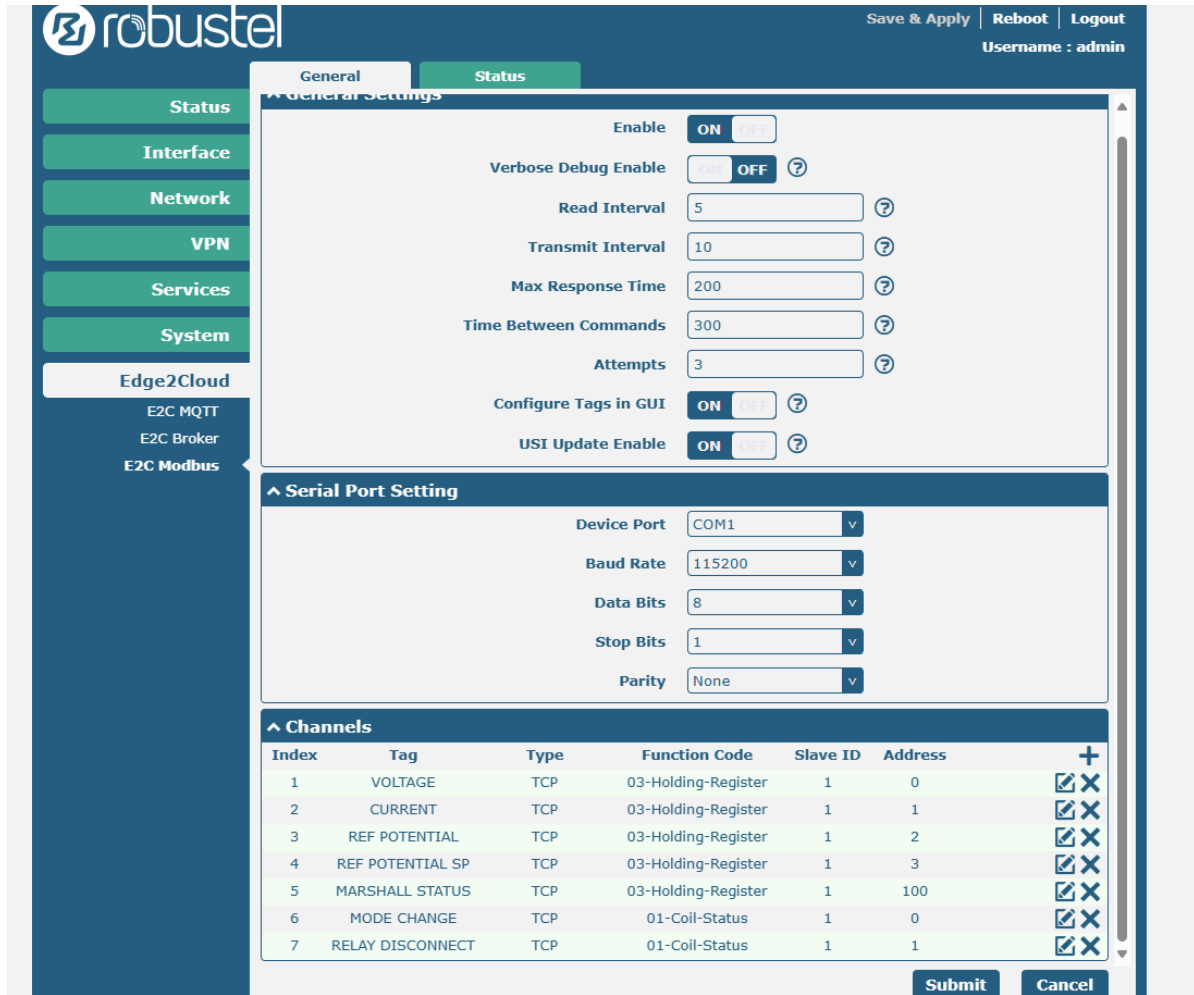
The bottom of the interface shows the copyright notice: 'Copyright © 2025 Robustel Technologies. All rights reserved.'

- 2.8. If the router profile has not been setup yet, then the global router template can be applied. Click on System > Profile. Then click on Choose file to import. Then browse in the xml configuration file. Click on Import. Save & apply the configuration.



The screenshot shows the Robustel web interface. The top navigation bar includes the Robustel logo, a 'Save & Apply' button, a 'Reboot' button, and a 'Logout' button. The user is logged in as 'admin'. The left sidebar contains a menu with the following items: Status, Interface, Network, VPN, Services, System (with sub-items: Debug, Update, App Center, Tools), Profile (with sub-items: Access Control, User Management, Role Management), and Edge2Cloud. The main content area is titled 'Profile' and 'Rollback'. It contains three sections: 'Import Configuration File' with options to 'Reset Other Settings to Default' (ON/OFF), 'Ignore Invalid Settings' (ON/OFF), and an 'XML Configuration File' field with a 'Choose File' button and an 'Import' button; 'Export Configuration File' with options to 'Ignore Disabled Features' (ON/OFF) and 'Add Detailed Information' (ON/OFF), and an 'XML Configuration File' field with a 'Generate' button; and 'Default Configuration' with a 'Save Running Configuration as Default' button and a 'Restore to Default Configuration' button. The footer of the interface states 'Copyright © 2025 Robustel Technologies. All rights reserved.'

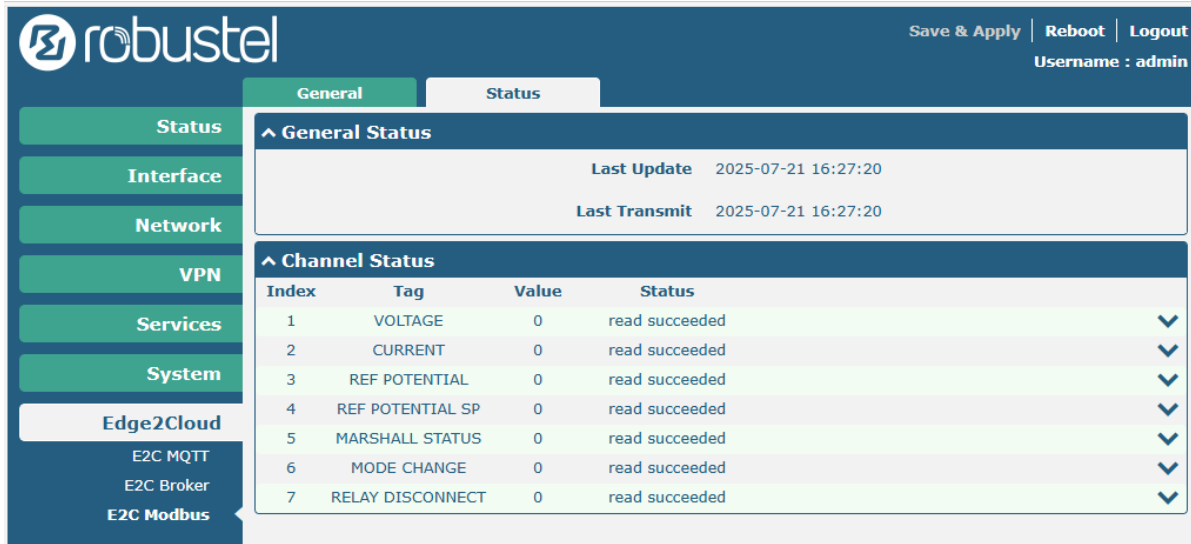
2.9. Click on the Edge2Cloud Section and select Modbus. See that the Modbus client is enabled and that the Modbus pointing table is applied.



The screenshot shows the Robustel Edge2Cloud web interface. The left sidebar contains navigation links: Status, Interface, Network, VPN, Services, System, and Edge2Cloud. The Edge2Cloud section is expanded, showing E2C MQTT, E2C Broker, and E2C Modbus. The main content area is titled 'General Settings' and has two tabs: General and Status. The Status tab is selected. The General Settings section includes a list of configuration options with toggle switches and input fields. Below this is the 'Serial Port Setting' section with dropdown menus for Device Port, Baud Rate, Data Bits, Stop Bits, and Parity. At the bottom is the 'Channels' section, which contains a table of data points.

Index	Tag	Type	Function Code	Slave ID	Address	
1	VOLTAGE	TCP	03-Holding-Register	1	0	☒ ☐
2	CURRENT	TCP	03-Holding-Register	1	1	☒ ☐
3	REF POTENTIAL	TCP	03-Holding-Register	1	2	☒ ☐
4	REF POTENTIAL SP	TCP	03-Holding-Register	1	3	☒ ☐
5	MARSHALL STATUS	TCP	03-Holding-Register	1	100	☒ ☐
6	MODE CHANGE	TCP	01-Coil-Status	1	0	☒ ☐
7	RELAY DISCONNECT	TCP	01-Coil-Status	1	1	☒ ☐

2.10. Click on Status and see that the router successfully read from the connected PLC.



robustel

Save & Apply | Reboot | Logout
Username : admin

General | Status

General Status

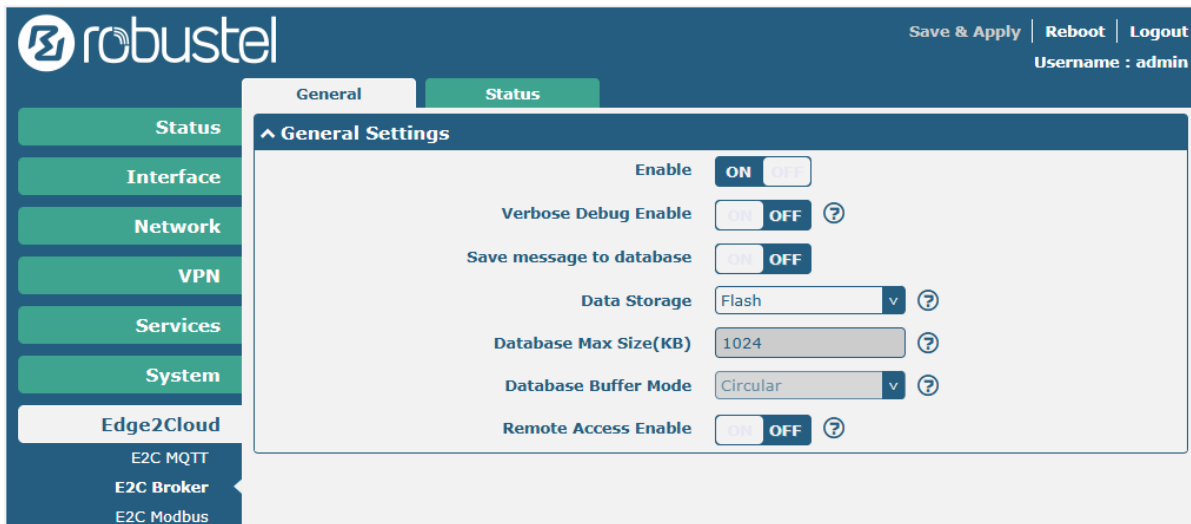
Last Update 2025-07-21 16:27:20
Last Transmit 2025-07-21 16:27:20

Channel Status

Index	Tag	Value	Status
1	VOLTAGE	0	read succeeded
2	CURRENT	0	read succeeded
3	REF POTENTIAL	0	read succeeded
4	REF POTENTIAL SP	0	read succeeded
5	MARSHALL STATUS	0	read succeeded
6	MODE CHANGE	0	read succeeded
7	RELAY DISCONNECT	0	read succeeded

Left sidebar: Status, Interface, Network, VPN, Services, System, Edge2Cloud, E2C MQTT, E2C Broker, E2C Modbus

2.11. Click on the E2C Broker and verify that the Modbus Broker Client is enabled.



robustel

Save & Apply | Reboot | Logout
Username : admin

General | Status

General Settings

Enable ☒ ON

Verbose Debug Enable ☐ OFF ?

Save message to database ☐ OFF

Data Storage Flash ?

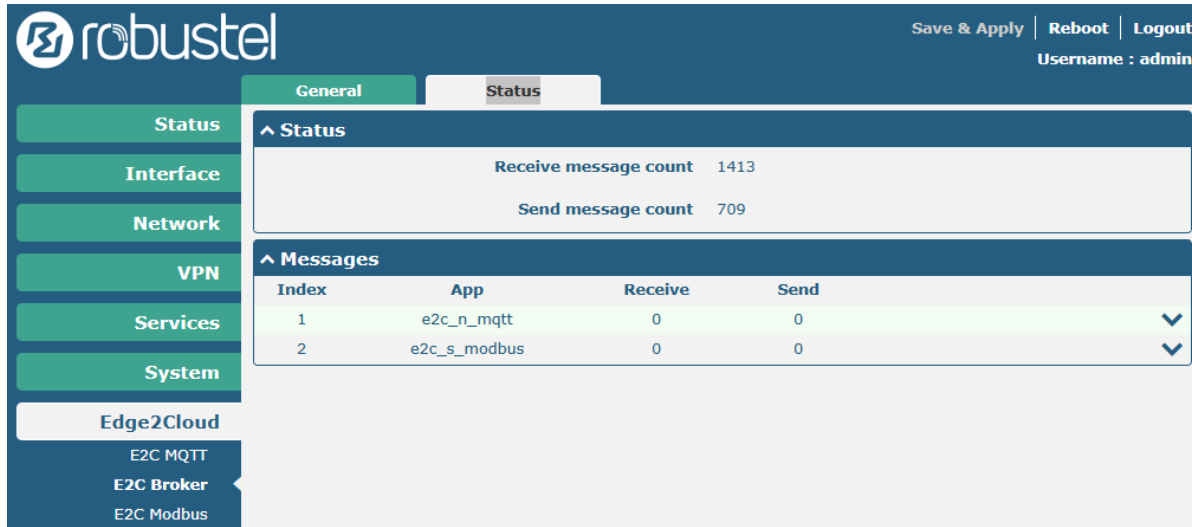
Database Max Size(KB) 1024 ?

Database Buffer Mode Circular ?

Remote Access Enable ☐ OFF ?

Left sidebar: Status, Interface, Network, VPN, Services, System, Edge2Cloud, E2C MQTT, E2C Broker, E2C Modbus

2.12. Click on status to ensure that Modbus is posting values to the broker.



The screenshot shows the Robustel Edge2Cloud web interface. The top navigation bar includes 'Save & Apply', 'Reboot', and 'Logout' buttons, along with the 'Username : admin' text. The left sidebar contains a menu with 'Status', 'Interface', 'Network', 'VPN', 'Services', 'System', and 'Edge2Cloud'. The 'Status' tab is selected, showing a 'Status' section with 'Receive message count' (1413) and 'Send message count' (709). Below this is a 'Messages' table with columns 'Index', 'App', 'Receive', and 'Send'. The table lists two entries: 'e2c_n_mqtt' and 'e2c_s_modbus', both with 0 receive and 0 send counts. The 'Edge2Cloud' section is expanded, showing 'E2C MQTT', 'E2C Broker', and 'E2C Modbus' options.

- 2.13. Click on E2C MQTT. Disable the client if not already disabled. Submit, Save & Apply.
- Ensure that the broker URL, username and password has been applied.
 - Copy device serial number to Client ID from notepad.
 - Apply the correct Topic name e.g. MIDM_TRU0003 to “/read, /response, /write, /status” topics. The global template default is “MIDM_TRU0000”.
 - Submit, Save & Apply.



Save & Apply
Reboot
Logout
Username : admin

MQTT
Status
Cert Manager

Status
Interface
Network
VPN
Services
System
Edge2Cloud
E2C MQTT
E2C Broker
E2C Modbus

General Settings

Enable
☐ ON ☒ OFF

Verbose Debug Enable
☐ ON ☒ OFF ?

MQTT TLS Enable
☒ ON ☐ OFF

MQTT Retain Enable
☒ ON ☐ OFF

QOS
 v

Report Interval
 ?

Remote Server

Remote Port

Username

Password

Client Id

KeepAlive
 ?

Topic Settings

Publish Topic

Response Topic

Subscribe Topic

Birth Topic

Birth Message

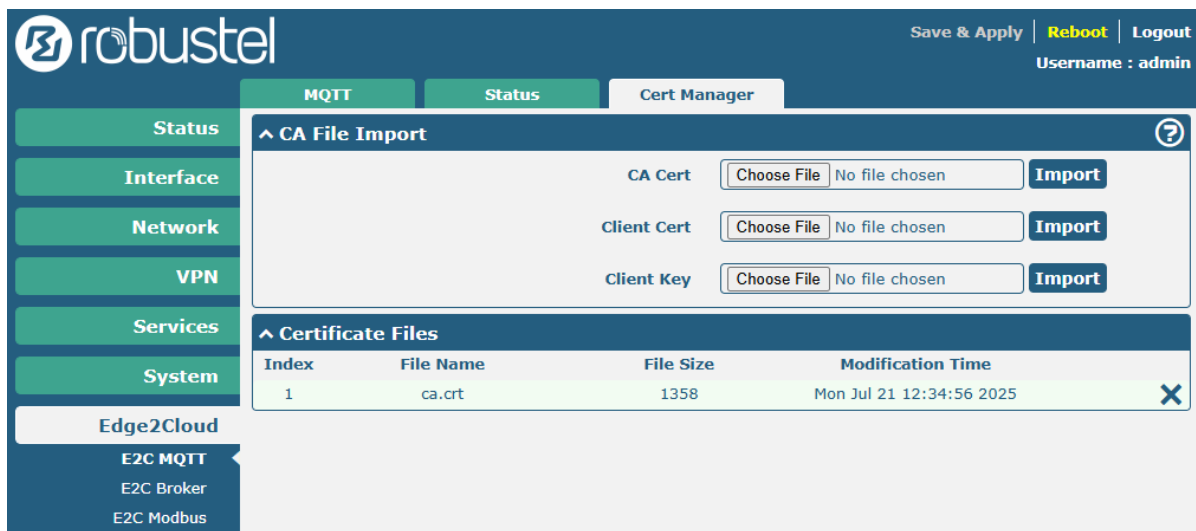
Last Will Topic

Last Will Message

Submit
Cancel

Copyright © 2025 Robustel Technologies. All rights reserved.

- 2.14. Click on Cert Manager and verify that the broker authorisation certificate has been applied. If not, then browse in the provided broker authorisation *.cert file. Submit, Save & Apply.



robustel

Save & Apply | **Reboot** | Logout
Username : admin

MQTT | Status | Cert Manager

CA File Import

CA Cert No file chosen

Client Cert No file chosen

Client Key No file chosen

^ Certificate Files

Index	File Name	File Size	Modification Time
1	ca.crt	1358	Mon Jul 21 12:34:56 2025

Edge2Cloud

- E2C MQTT
- E2C Broker
- E2C Modbus

2.15. Click on E2C MQTT. Enable the broker. Submit, Save & Apply.



Save & Apply | Reboot | Logout
Username : admin

MQTT | Status | Cert Manager

Status
Interface
Network
VPN
Services
System
Edge2Cloud
E2C MQTT
E2C Broker
E2C Modbus

General Settings

Enable ☒ ON ☐ OFF

Verbose Debug Enable ☐ OFF ?

MQTT TLS Enable ☒ ON ☐ OFF

MQTT Retain Enable ☒ ON ☐ OFF

QOS v

Report Interval ?

Remote Server

Remote Port

Username

Password

Client Id

KeepAlive ?

Topic Settings

Publish Topic

Response Topic

Subscribe Topic

Birth Topic

Birth Message

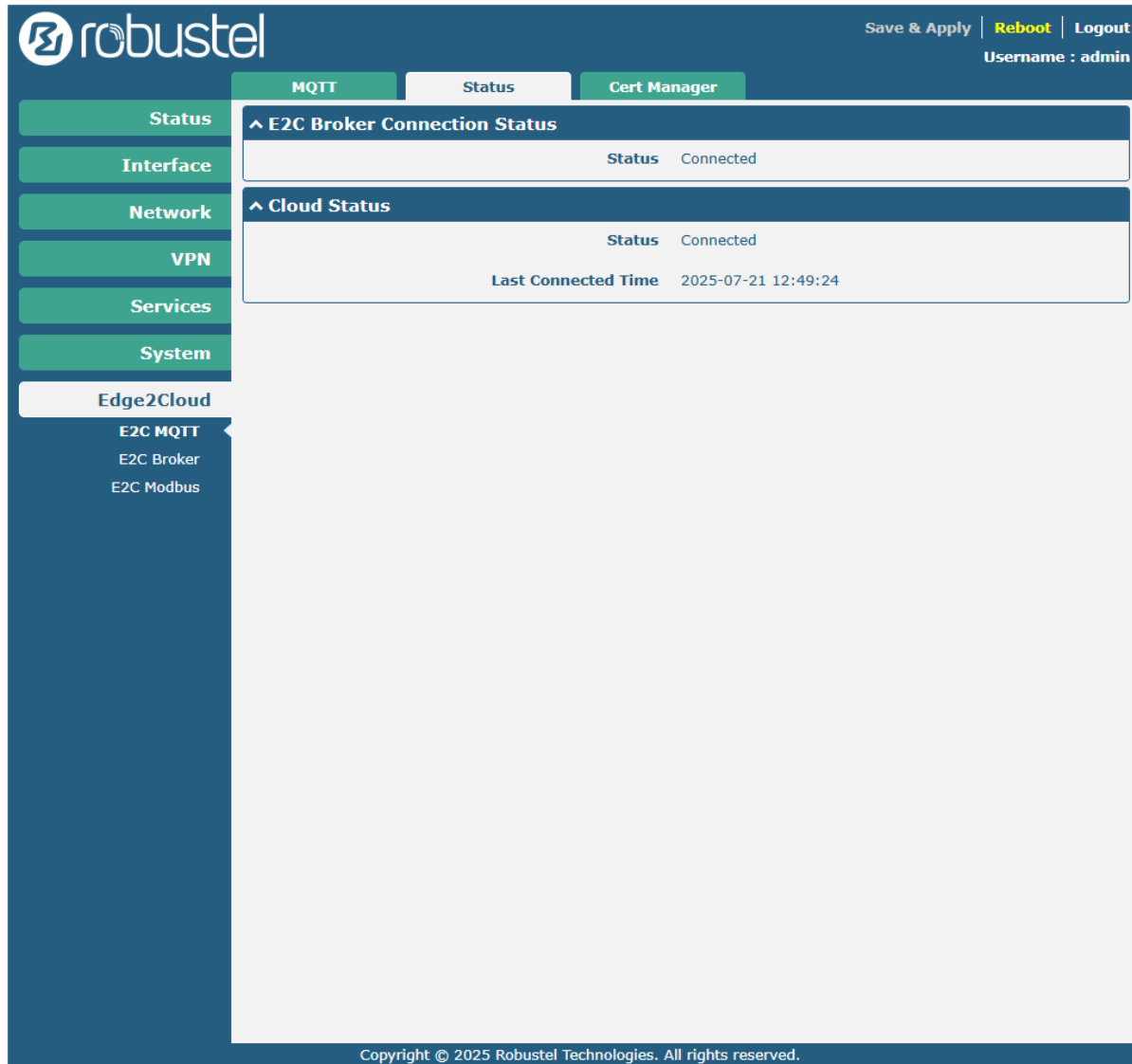
Last Will Topic

Last Will Message

Submit Cancel

Copyright © 2025 Robustel Technologies. All rights reserved.

2.16. Click on E2C MQTT Status. The router should now be connected to the specified broker.



The screenshot shows the Robustel web interface. The top header includes the Robustel logo, navigation links (Save & Apply, Reboot, Logout), and the username (admin). The left sidebar contains a menu with options: Status, Interface, Network, VPN, Services, System, Edge2Cloud, E2C MQTT, E2C Broker, and E2C Modbus. The main content area displays the MQTT status, showing E2C Broker Connection Status as Connected and Cloud Status as Connected, with a last connected time of 2025-07-21 12:49:24. The footer contains the copyright notice: Copyright © 2025 Robustel Technologies. All rights reserved.

- 2.17. Click on Services and then NTP (Network Time Protocol). Ensure that the Time zone is set to local UTC time. The NTP Client must be enabled. Add server 0.za.pool.ntp.org as a secondary NTP server and change NTP Update Interval to 10 min.



[Save & Apply](#) | [Reboot](#) | [Logout](#)
Username : admin

Status

Interface

Network

VPN

Services

System

Edge2Cloud

Syslog

Event

NTP

SMS

Email

DDNS

SSH

RCMS

Web Server

Advanced

Smart Roaming V2

NTP

Status

Timezone Settings

Time Zone

UTC+02:00

▼

Expert Setting

?

NTP Client Settings

Enable

ON

OFF

Primary NTP Server

pool.ntp.org

?

Secondary NTP Server

server 0.za.pool.ntp.or

?

NTP Update Interval

10

?

Request network port

default

▼

NTP Server Settings

Enable

ON

OFF

2.18. Click on Network, Firewall and verify that the Firewall is enabled. Reboot router if needed or logout when finished.



[Save & Apply](#)
[Reboot](#)
[Logout](#)

Username : admin

Status

Interface

Network

Firewall

VPN

Services

System

Edge2Cloud

Route

QoS

IP Passthrough

PPPoE Bridge

Filtering

NAT

Advanced

Custom Rules

Status

General Settings

Enable Filtering

ON

OFF

Default Filtering Policy

Drop

?

Remote Input Policy

Accept

?

Local Input Policy

Accept

?

Access Control Settings

Enable Remote SSH Access

ON

OFF

Enable Local SSH Access

ON

OFF

Enable Remote Telnet Access

ON

OFF

Enable Local Telnet Access

ON

OFF

Enable Remote HTTP Access

ON

OFF

Enable Local HTTP Access

ON

OFF

Enable Remote HTTPS Access

ON

OFF

Enable Remote Ping Respond

ON

OFF

?

Enable DOS Defending

ON

OFF

Enable VPN NAT Traversal

ON

OFF

?

Enable Ping Forward Limit

ON

OFF

?

Whitelist Rules

Index

Description

Source Address

+

Filtering Rules

Index

Source Address

Source Port

Source MAC

Target Address

Target Port

Protocol

+

MAC IP Binding

Index

MAC Address

IP Address

+

Submit

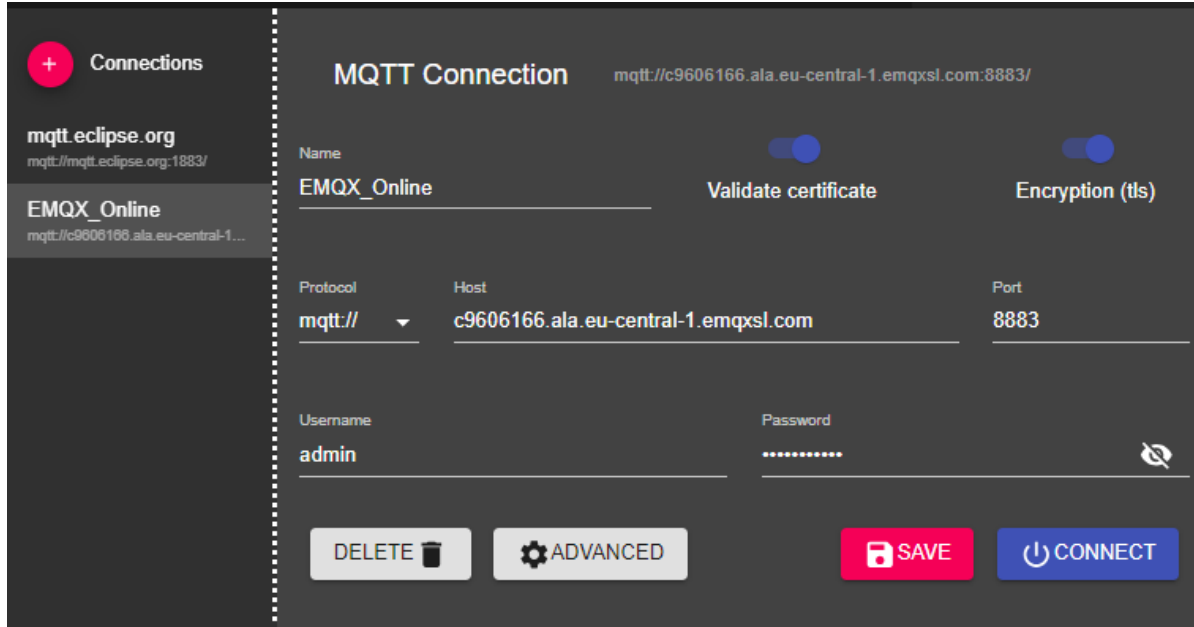
Cancel

Copyright © 2025 Robustel Technologies. All rights reserved.

16

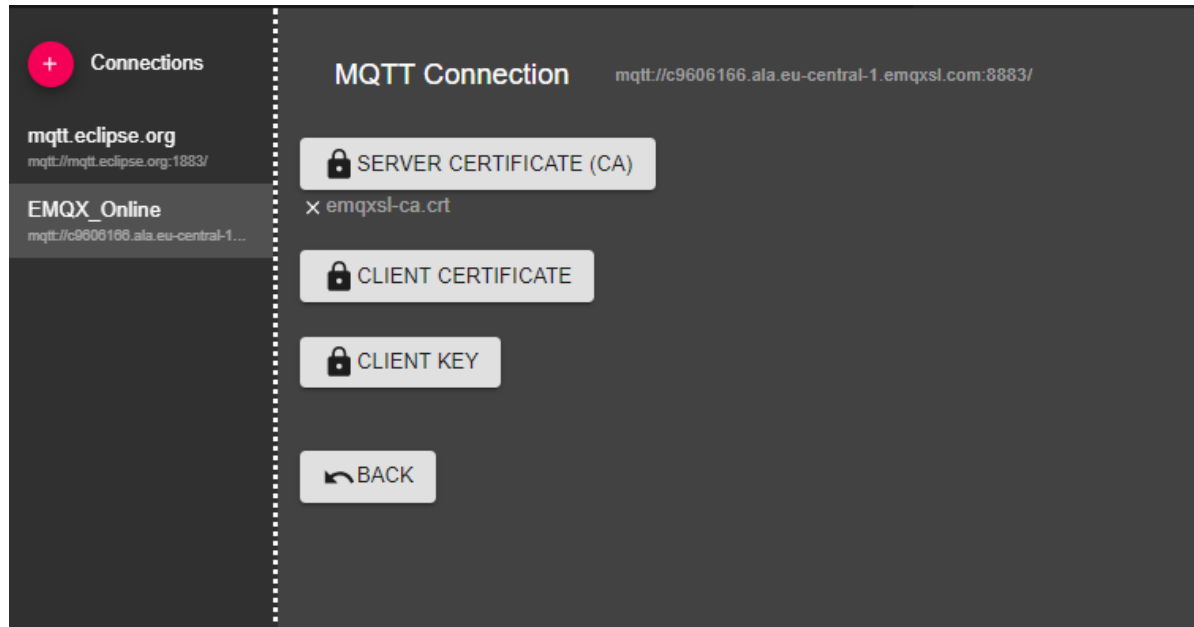
3. MQTT Explorer Setup & Testing

3.1. Open MQTT Explorer. Ensure that you've configured the provided broker connection, username and password details.

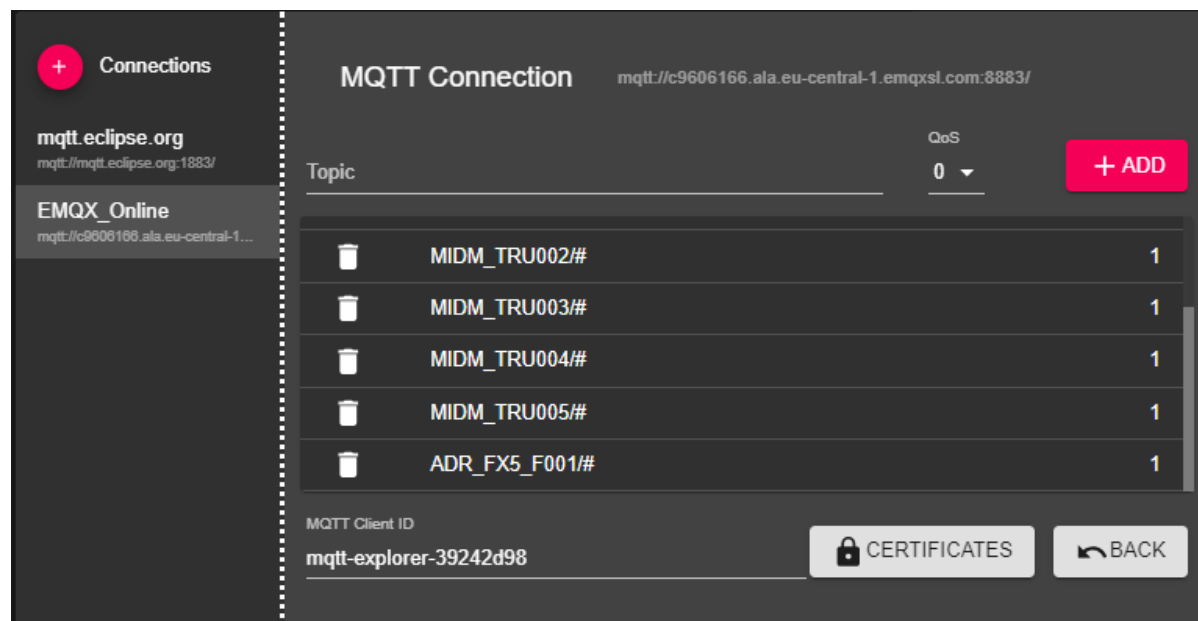


The screenshot displays the MQTT Explorer application interface. On the left, a sidebar shows a list of connections: 'mqtt.eclipse.org' and 'EMQX_Online'. The 'EMQX_Online' connection is selected. The main area is titled 'MQTT Connection' and shows the connection string 'mqtt://c9606166.ala.eu-central-1.emqxsl.com:8883/'. Below this, there are fields for 'Name' (EMQX_Online), 'Validate certificate' (checked), and 'Encryption (tls)' (checked). The 'Protocol' is set to 'mqtt://', the 'Host' is 'c9606166.ala.eu-central-1.emqxsl.com', and the 'Port' is '8883'. The 'Username' is 'admin' and the 'Password' is masked with dots. At the bottom, there are buttons for 'DELETE', 'ADVANCED', 'SAVE', and 'CONNECT'.

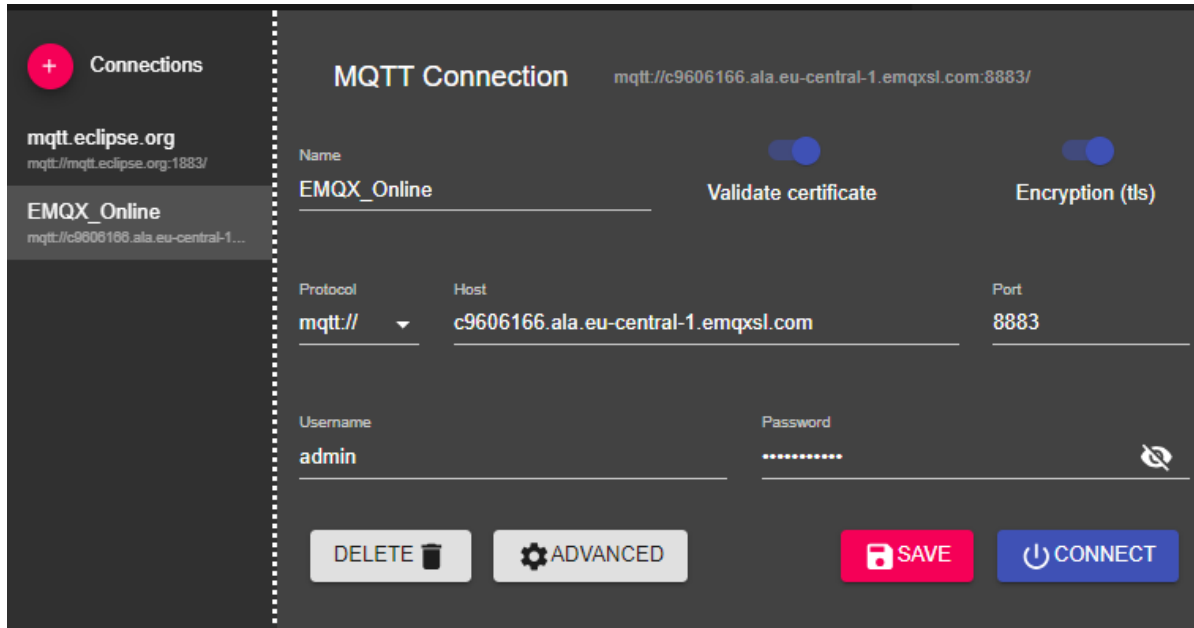
3.2. Click on certificate and ensure that the broker authorisation certificate is applied.



- 3.4. Click on Back and make sure that your device topic & QOS has been applied for subscription. It must be the same as what you have configured under step 13.

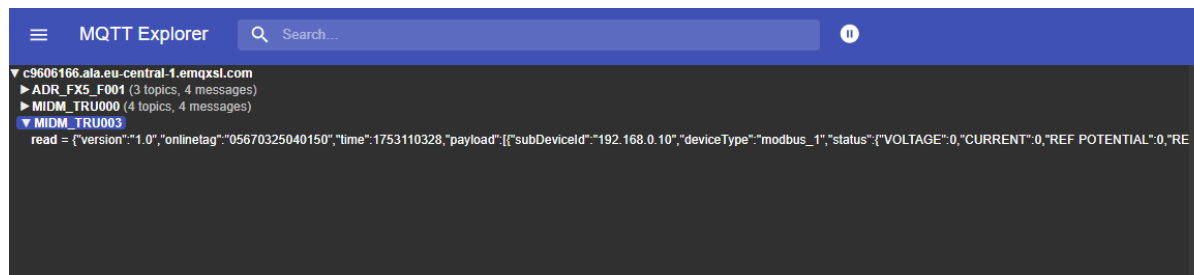


3.5. Click on back, making sure that that you click on save to apply your configuration. Click on Connect.

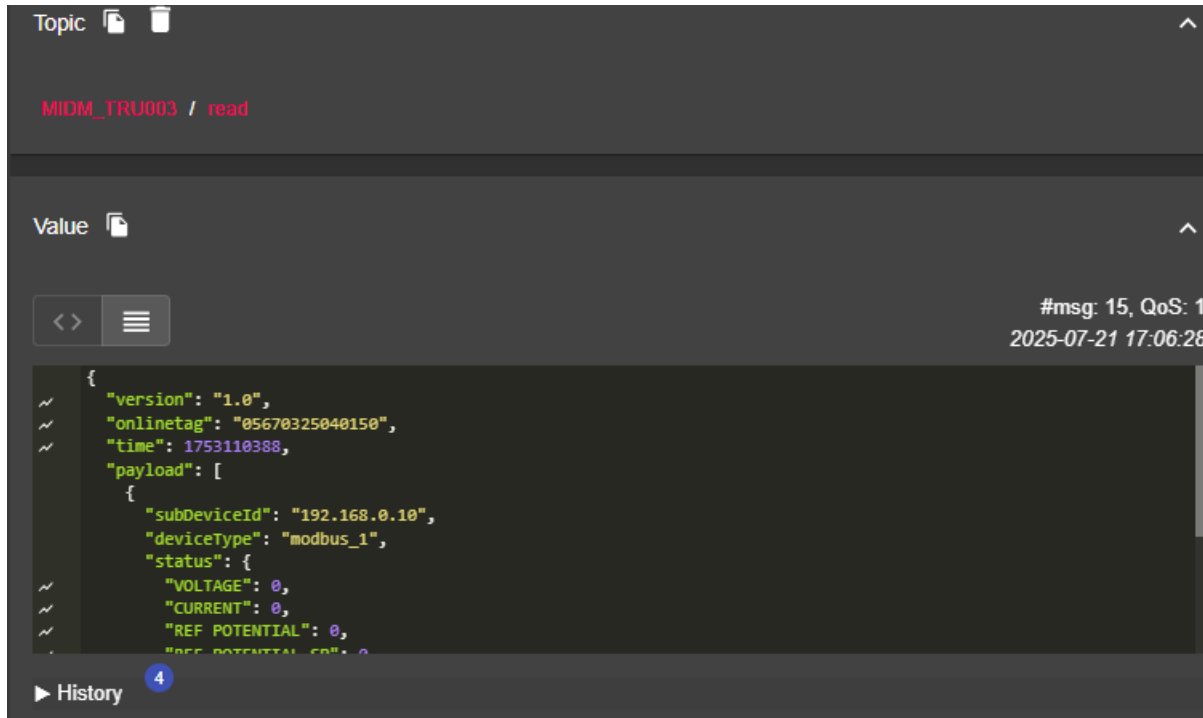


The image shows the MQTT Connection configuration interface. On the left, there is a sidebar with a 'Connections' section containing a list of connections: 'mqtt.eclipse.org' and 'EMQX_Online'. The 'EMQX_Online' connection is selected. The main area is titled 'MQTT Connection' and shows the connection URL 'mqtt://c9606166.ala.eu-central-1.emqxsl.com:8883/'. Below this, there are fields for 'Name' (EMQX_Online), 'Validate certificate' (checked), and 'Encryption (tls)' (checked). There are also fields for 'Protocol' (mqtt://), 'Host' (c9606166.ala.eu-central-1.emqxsl.com), and 'Port' (8883). The 'Username' field is set to 'admin' and the 'Password' field is masked with dots. At the bottom, there are buttons for 'DELETE', 'ADVANCED', 'SAVE', and 'CONNECT'.

3.6. The device MQTT posted topic should appear in the explorer.



3.7. Click on the read topic. You should see the values posted from the PLC registers along with device serial number & UNIX timestamp. You can use [Unix Time Stamp - Epoch Converter](#) to check accuracy on timestamp received from the device.



```
Topic MIDM_TRU003 / read

Value

#msg: 15, QoS: 1
2025-07-21 17:06:28

{
  "version": "1.0",
  "onlinetag": "05670325040150",
  "time": 1753110388,
  "payload": [
    {
      "subDeviceId": "192.168.0.10",
      "deviceType": "modbus_1",
      "status": {
        "VOLTAGE": 0,
        "CURRENT": 0,
        "REF POTENTIAL": 0,
        "REF POTENTIAL CO": 0
      }
    }
  ]
}
```

History 4

3.8. You can “test” publish a value to the Modbus device and monitor the update back.

Example:

Add the corresponding device topic to the string and change the value as needed. In the below example we’ll write a value to the topic **"REF POTENTIAL SP"** and verify the published update under the device read topic.

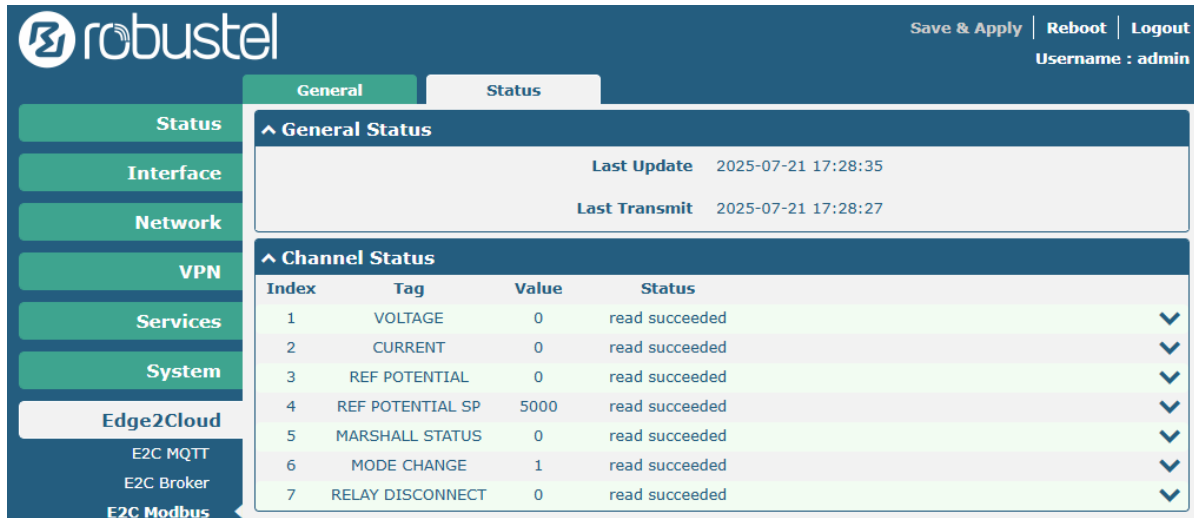
Topic:
MIDM_TRU003/write

Payload:

```
{  
  "payload": {  
    "subDeviceId": "192.168.0.10",  
    "subProtocol": "modbus",  
    "subDeviceTag": "1",  
    "cmdType": "writeReg",  
    "param": {  
      "REF POTENTIAL SP": 5000  
    }  
  }  
}
```



3.9. Device is writing to the PLC and is updating to the broker successfully.



The screenshot shows the Robustel Edge2Cloud web interface. The top navigation bar includes 'Save & Apply', 'Reboot', and 'Logout' buttons, along with the 'Username : admin' field. The left sidebar contains a menu with 'Status', 'Interface', 'Network', 'VPN', 'Services', 'System', and 'Edge2Cloud' (which is expanded to show 'E2C MQTT', 'E2C Broker', and 'E2C Modbus'). The main content area is divided into two tabs: 'General' and 'Status'. The 'Status' tab is active, displaying the 'General Status' section with 'Last Update' and 'Last Transmit' timestamps. Below this is the 'Channel Status' section, which contains a table with 7 rows of data.

Index	Tag	Value	Status
1	VOLTAGE	0	read succeeded
2	CURRENT	0	read succeeded
3	REF POTENTIAL	0	read succeeded
4	REF POTENTIAL SP	5000	read succeeded
5	MARSHALL STATUS	0	read succeeded
6	MODE CHANGE	1	read succeeded
7	RELAY DISCONNECT	0	read succeeded

4. Adroit Configuration

4.1. Installation

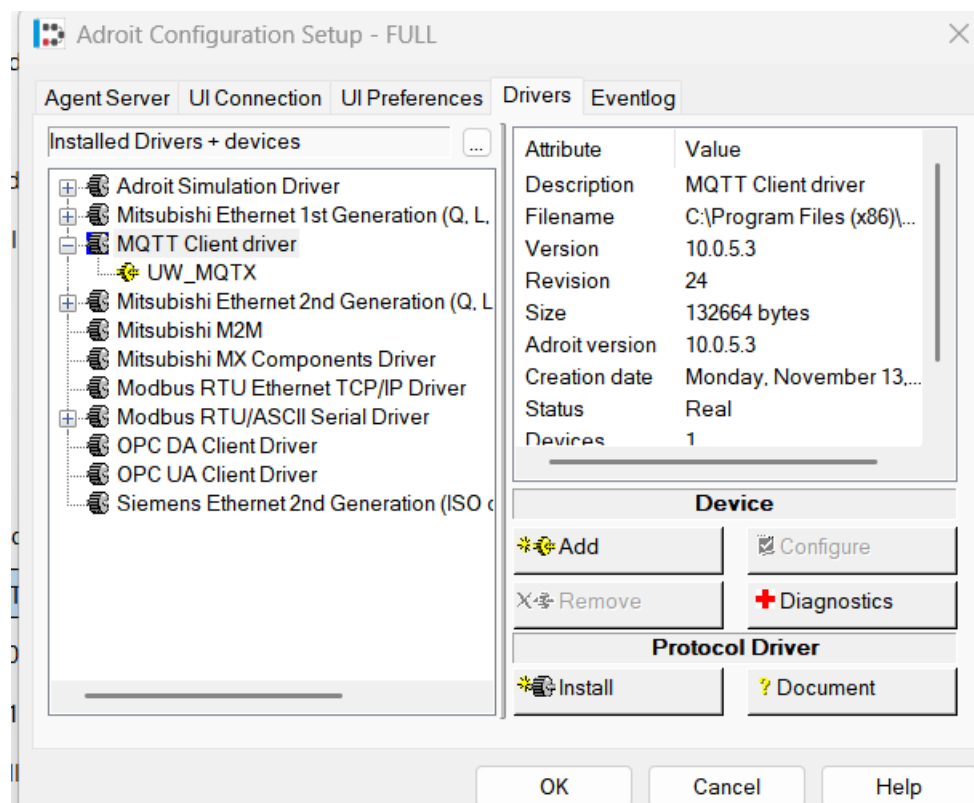
- Ensure running the latest and official release of Adroit
- Go to C:\Program Files (x86)\Adroit Technologies\Adroit and verify
 - o d_mqtt_cl.dll rev.31 or higher
 - o x_iot.dll rev.32 or higher

4.2. Copy x_iot.dll and rename the copy created to your IOT and Type name as needed.

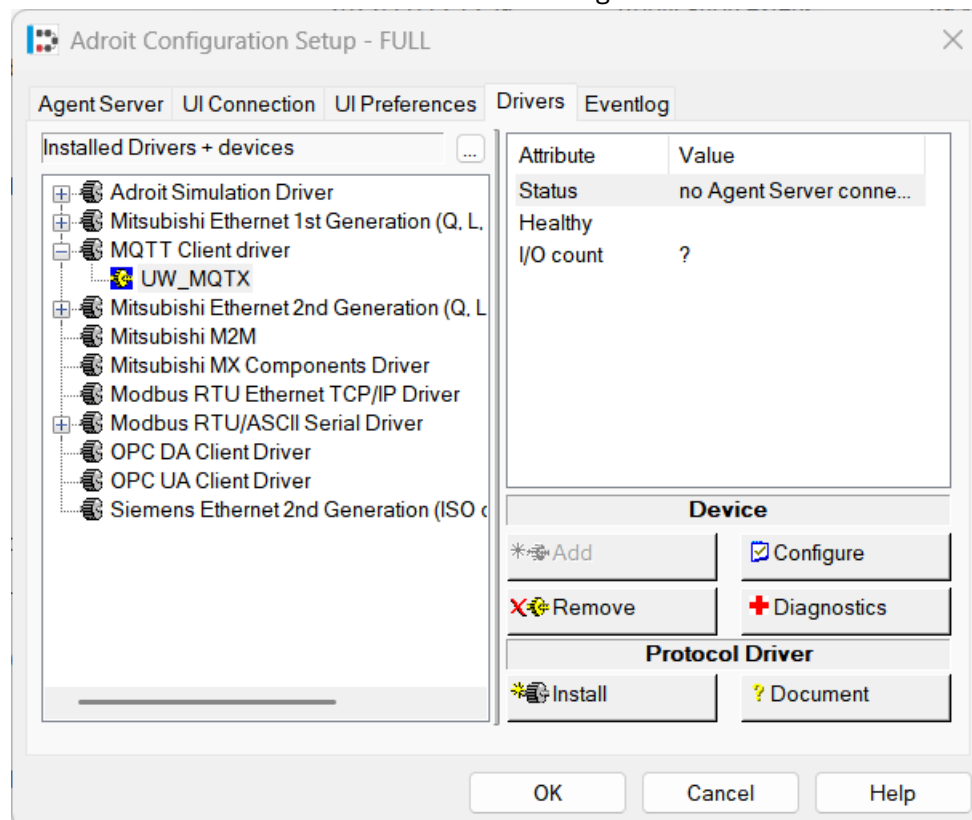
Example: t_MQTT_FX5U_T1.dll.

Different IoT devices with different I/O and topics will warrant its own configuration. You will then have a *.dll for that device as well.

4.3. Run the Adroit Configuration Setup and Install the MQTT driver or add the Adroit MQTT client. Clients connect to a single broker or API's with a secondary connection should the primary connection not be available.



4.4. Select the added device and click on configure.



4.5. Configure the client with the Broker authentication details. Store the Broker certificate *.crt file locally and copy the full path to the .crt file into CA path. Best to store this file as part of the Adroit project configuration files. For more detail and help, please refer to the MQTT driver documentation.

MQTT Client driver - UW_MQTX

☐ Enable secondary ☒ Primary ☐ Secondary

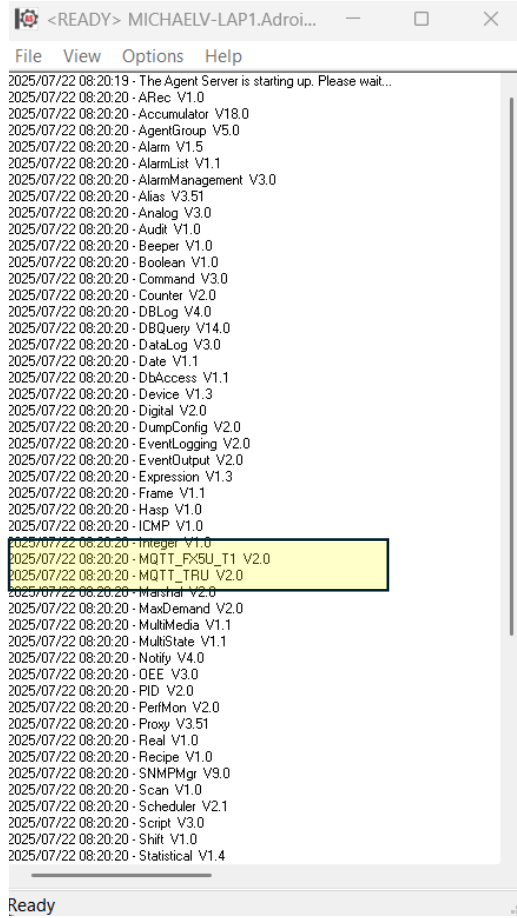
Primary connection parameters

☐ Enable shadow scanning from master to standby ☐ Enable standby reads
☐ Enable standby writes

Property	Value
Client details	
Client name	8QKeF0oxUZbs62ZIFpzCiwD
Broker details	
Broker host IP	c9606166.ala.eu-central-1.emqxsl.com
Broker port	8883
Clean session	False
Retain	False
QoS	At least once
Keep alive (sec)	60
Transaction timeout (sec)	10
Connection timeout (sec)	10
Topic prefix	
Security	
SSL/TLS mode	True
TLS version (fixed)	12
CA path (future)	C:\MosquittoSSL\certs\emqxsl-ca.crt
Certificate (future)	
User name	admin
User password	*****
Last will	
Will topic	AdroitSCADA/status
Will message	"Scada Offline"
Will QoS	At most once
Will retain	False
Firewall	
Proxy type	None
Proxy auto detect	False
Proxy hostname	
Proxy username	
Proxy password	
Proxy port	0
Test	
Ping size	32
Broker ping	-

Close

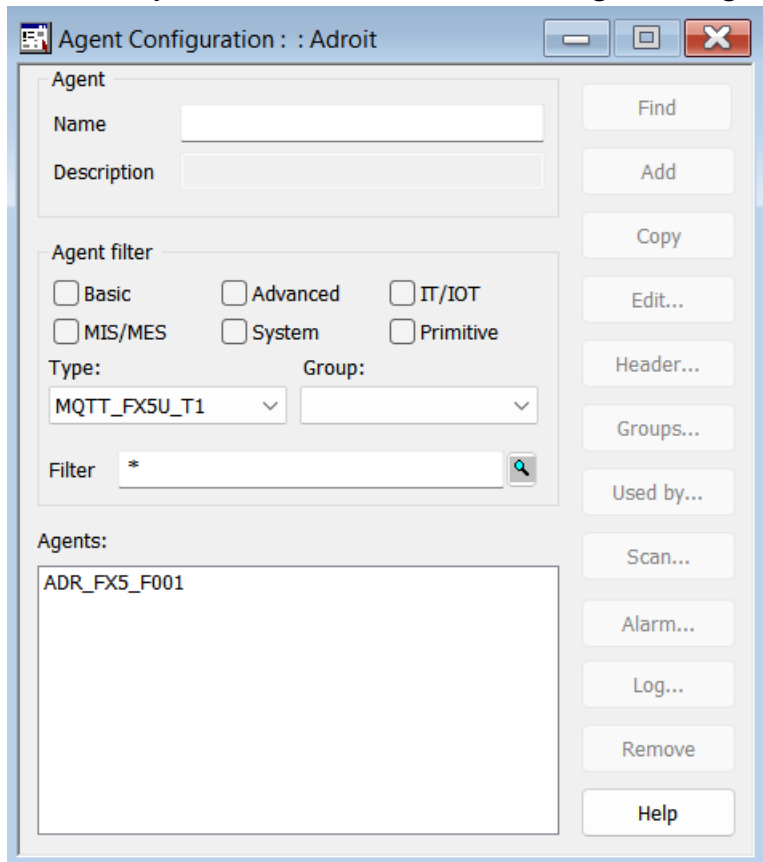
4.6. Start the Adroit Agent Server. See that the Agent Server starts-up and that the Agents and IOT devices created loads without any errors.



```

2025/07/22 08:20:19 - The Agent Server is starting up. Please wait...
2025/07/22 08:20:20 - ARec V1.0
2025/07/22 08:20:20 - Accumulator V18.0
2025/07/22 08:20:20 - AgentGroup V5.0
2025/07/22 08:20:20 - Alarm V1.5
2025/07/22 08:20:20 - AlarmList V1.1
2025/07/22 08:20:20 - AlarmManagement V3.0
2025/07/22 08:20:20 - Alias V3.51
2025/07/22 08:20:20 - Analog V3.0
2025/07/22 08:20:20 - Audit V1.0
2025/07/22 08:20:20 - Beeper V1.0
2025/07/22 08:20:20 - Boolean V1.0
2025/07/22 08:20:20 - Command V3.0
2025/07/22 08:20:20 - Counter V2.0
2025/07/22 08:20:20 - DBLog V4.0
2025/07/22 08:20:20 - DBQuery V14.0
2025/07/22 08:20:20 - DataLog V3.0
2025/07/22 08:20:20 - Date V1.1
2025/07/22 08:20:20 - DbAccess V1.1
2025/07/22 08:20:20 - Device V1.3
2025/07/22 08:20:20 - Digital V2.0
2025/07/22 08:20:20 - DumpConfig V2.0
2025/07/22 08:20:20 - EventLogging V2.0
2025/07/22 08:20:20 - EventOutput V2.0
2025/07/22 08:20:20 - Expression V1.3
2025/07/22 08:20:20 - Frame V1.1
2025/07/22 08:20:20 - Hasp V1.0
2025/07/22 08:20:20 - ICMP V1.0
2025/07/22 08:20:20 - Integer V1.0
2025/07/22 08:20:20 - MQTT_FX5U_T1 V2.0
2025/07/22 08:20:20 - MQTT_TRU V2.0
2025/07/22 08:20:20 - Marshal V2.0
2025/07/22 08:20:20 - MaxDemand V2.0
2025/07/22 08:20:20 - MultiMedia V1.1
2025/07/22 08:20:20 - MultiState V1.1
2025/07/22 08:20:20 - Notify V4.0
2025/07/22 08:20:20 - OEE V3.0
2025/07/22 08:20:20 - PID V2.0
2025/07/22 08:20:20 - PerfMon V2.0
2025/07/22 08:20:20 - Proxy V3.51
2025/07/22 08:20:20 - Real V1.0
2025/07/22 08:20:20 - Recipe V1.0
2025/07/22 08:20:20 - SNMPMgr V9.0
2025/07/22 08:20:20 - Scan V1.0
2025/07/22 08:20:20 - Scheduler V2.1
2025/07/22 08:20:20 - Script V3.0
2025/07/22 08:20:20 - Shift V1.0
2025/07/22 08:20:20 - Statistical V1.4
  
```

4.6. Create your IoT device instance in Adroit Agent Configuration and scan it to the MQTT Client Device.



The screenshot shows the 'Agent Configuration' window in the Adroit software. The window has a title bar with the text 'Agent Configuration : : Adroit'. The main area is divided into several sections:

- Agent**: Contains text boxes for 'Name' and 'Description'.
- Agent filter**: Contains checkboxes for 'Basic', 'Advanced', 'IT/IOT', 'MIS/MES', 'System', and 'Primitive'. Below these are dropdown menus for 'Type' (set to 'MQTT_FX5U_T1') and 'Group'.
- Filter**: A text box with an asterisk '*' and a search icon.
- Agents:**: A list box containing the text 'ADR_FX5_F001'.

On the right side of the window, there is a vertical column of buttons: 'Find', 'Add', 'Copy', 'Edit...', 'Header...', 'Groups...', 'Used by...', 'Scan...', 'Alarm...', 'Log...', 'Remove', and 'Help'.

Agent Configuration : : Adroit

Tag scanning configuration : UW_MQTX

Device: UW_MQTX ☒ Started ☐ Stopped

Agent: ADR_FX5_F001 ... Find

Slot: ☐ Header slots
rawValue

Address: S:ADR_FX5_F001/read ... Find

Scan rate (ms): 1000 Deadband: 0

☐ Input disabled ☐ Output enabled
☐ Demand scan ☐ Output initialize

Currently scanned tags:

Tag	Address	Rate	Deadband	IO flags
ADR_FX5_F001.rawValue	S:ADR_FX5_F001/read	1000	0	1

Scan Unscan

OK Help Details... Import...

4.7. On a successful payload read from the Broker, the IoT Agent will populate the “headerslots” from the topics received. You can export the Agent to an .xml file where you can further add slots and expression to customise the Agent as needed. Once the templates is applied, save your configuration, and restart Adroit UI and Adroit Agent Server before adding more IoT device instances. The template and slots will now be universally applied to all instance of the same type.

 Edit IOT agent : ADR_FX5_F001

Slot information:

Name	Description	Type	Routing agent type	Value
version	from JSON	String		1.0
onlinetag	from JSON	String		05670325040173
time	from JSON	Integer		1753165895
payload1subDeviceId	from JSON	String		192.168.0.10
payload1deviceType	from JSON	String		modbus_1
DIN_X0_17	from JSON	Integer		0
DOT_Y0_17	from JSON	Integer		0
AIN_D0	from JSON	Integer		0
AIN_D1	from JSON	Integer		18042
AIN_D2	from JSON	Integer		0
AIN_D3	from JSON	Integer		0
AOT_D10	from JSON	Integer		16000

Expression:

OK

Cancel

Update

Refresh

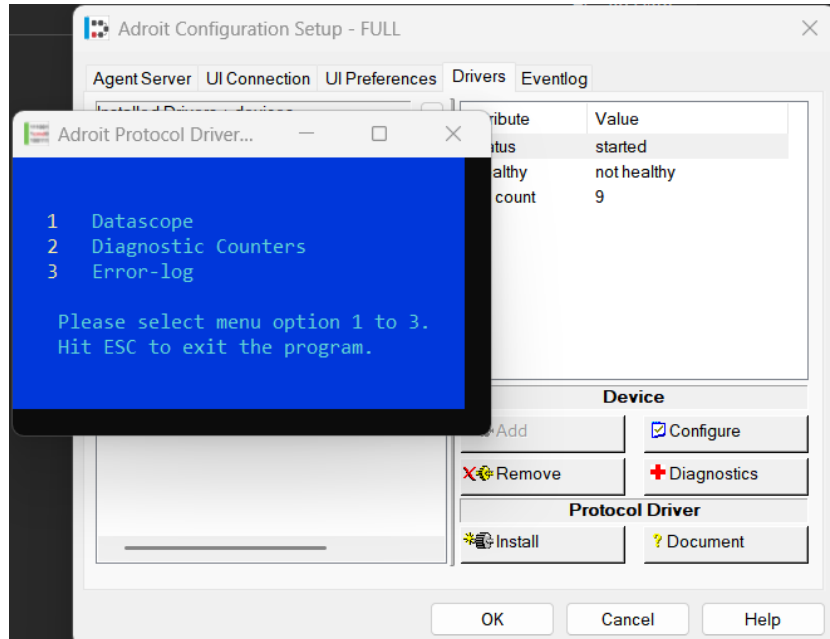
Header

Help

Load template

Save template

4.8. From Adroit Configuration Setup, select the MQTT device and launch the Datascope (Press 1)



4.9. Monitor for */read payload messages received from the Broker.

```

ASCII DATASCOPE for UW_MQTX on Pri,(A) Address c9606166.ala.eu-central-1.emqxsl.com:8883

Task = 0xBC5EE60 total time=65, (m_dwTicks_Allocated = 1096049, m_dwTicks_Build = 1096114, m_dwTicks_Transact = 790271, m_dwTicks_DataReceived = 0, m_dwTicks_UpdateStarted = 0, m_dwTicks_UpdateEnded = 1083894, m_dwTicks_Deallocated = 1096114)

Additional Log Info [[MQTT] Incoming packet type (3)PUBLISH.
00000000 32 BA 02 00 11 41 44 52 5F 46 58 35 5F 46 30 30 2...ADR_FX5_F00
00000010 31 2F 72 65 61 64 00 26 7B 22 76 65 72 73 69 6F 1/read.&{"versio
00000020 6E 22 3A 22 31 2E 30 22 2C 22 6F 6E 6C 69 6E 65 n":"1.0","online
00000030 74 61 67 22 3A 22 30 35 36 37 30 33 32 35 30 34 tag":"0567032504
00000040 30 31 37 33 22 2C 22 74 69 6D 65 22 3A 31 37 35 0173","time":175
00000050 33 31 36 35 35 35 34 2C 22 70 61 79 6C 6F 61 64 3165554,"payload
00000060 22 3A 5B 7B 22 73 75 62 44 65 76 69 63 65 49 64 ":[{"subDeviceId
00000070 22 3A 22 31 39 32 2E 31 36 38 2E 30 2E 31 30 22 ": "192.168.0.10"
00000080 2C 22 64 65 76 69 63 65 54 79 70 65 22 3A 22 6D ,"deviceType":"m
00000090 6F 64 62 75 73 5F 31 22 2C 22 73 74 61 74 75 73 odbus_1","status
000000A0 22 3A 7B 22 44 49 4E 5F 58 30 5F 31 37 22 3A 30 ":{"DIN_X0_17":0
000000B0 2C 22 44 4F 54 5F 59 30 5F 31 37 22 3A 30 2C 22 ,"DOT_Y0_17":0,"
000000C0 41 49 4E 5F 44 30 22 3A 30 2C 22 41 49 4E 5F 44 AIN_D0":0,"AIN_D
000000D0 31 22 3A 31 38 30 34 32 2C 22 41 49 4E 5F 44 32 1":18042,"AIN_D2
000000E0 22 3A 30 2C 22 41 49 4E 5F 44 33 22 3A 30 2C 22 ":0,"AIN_D3":0,"
000000F0 41 4F 54 5F 44 31 30 22 3A 31 36 30 30 30 2C 22 AOT_D10":16000,"
00000100 41 4F 54 5F 44 31 31 22 3A 30 2C 22 41 4F 54 5F AOT_D11":0,"AOT_
00000110 44 31 32 22 3A 30 2C 22 41 4F 54 5F 44 31 33 22 D12":0,"AOT_D13"
00000120 3A 30 2C 22 74 69 6D 65 73 74 61 6D 70 22 3A 31 :0,"timestamp":1
00000130 37 35 33 30 32 31 30 33 34 7D 7D 5D 7D 753021034}}]] ]
***** FireReadyToSend *****
Additional Log Info [[MQTT] Outgoing packet type (4)PUBACK.
00000000 40 02 00 26 @..& ]
DoFireMessageAck, PacketID:[38] Dir:[1] Idx:[0] InCount[1]out[0]

DoFireMessageIn [{"version":"1.0","onlinetag":"05670325040173","time":1753165554,"payload":[{"subDeviceId":"192.168.0.10","deviceType":"modbus_1","status":{"DIN_X0_17":0,"DOT_Y0_17":0,"AIN_D0":0,"AIN_D1":18042,"AIN_D2":0,"AIN_D3":0,"AOT_D10":16000,"AOT_D11":0,"AOT_D12":0,"AOT_D13":0,"timestamp":1753021034}}]] on PacketId [38] for Topic:[ADR_FX5_F001/read] QOS:[1] InCount[0]out[0],10

Addr= ADR_FX5_F001/read, Val= {"version":"1.0","onlinetag":"05670325040173","time":1753165554,"payload":[{"subDeviceId":"192.168.0.10","deviceType":"modbus_1","status":{"DIN_X0_17":0,"DOT_Y0_17":0,"AIN_D0":0,"AIN_D1":18042,"AIN_D2":0,"AIN_D3":0,"AOT_D10":16000,"AOT_D11":0,"AOT_D12":0,"AOT_D13":0,"timestamp":1753021034}}], PrevVal= {"version":"1.0","onlinetag":"05670325040173","time":1753165534,"payload":[{"subDeviceId":"192.168.0.10","deviceType":"modbus_1","status":{"DIN_X0_17":0,"DOT_Y0_17":0,"AIN_D0":0,"AIN_D1":18042,"AIN_D2":0,"AIN_D3":0,"AOT_D10":16000,"AOT_D11":0,"AOT_D12":0,"

```

4.10. Your Adroit MQTT device instance is now working. You can proceed to Aliascan your Agents to the device instance lots.

Tag scanning configuration : UW_MQTX

Device: ALIASCAN ☒ Started ☐ Stopped

Agent: ADR_FX5_F001_AI_D0 ... Find

Slot: ☐ Header slots
rawValue

Address: @ADR_FX5_F001.AIN_D0 ... Find

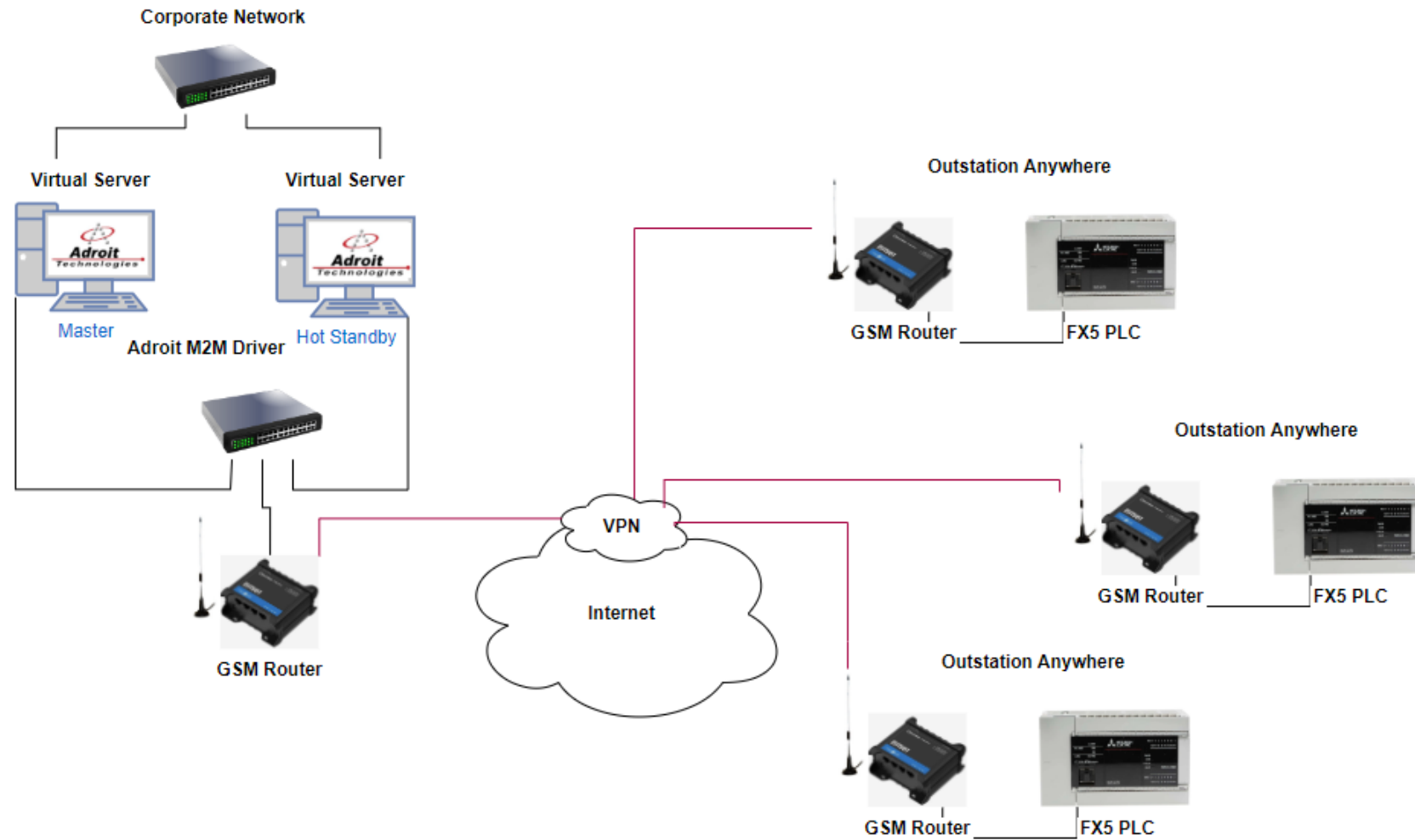
Scan rate (ms): 0 Deadband: 0

☐ Input disabled ☐ Output enabled
☐ Demand scan ☐ Output initialize

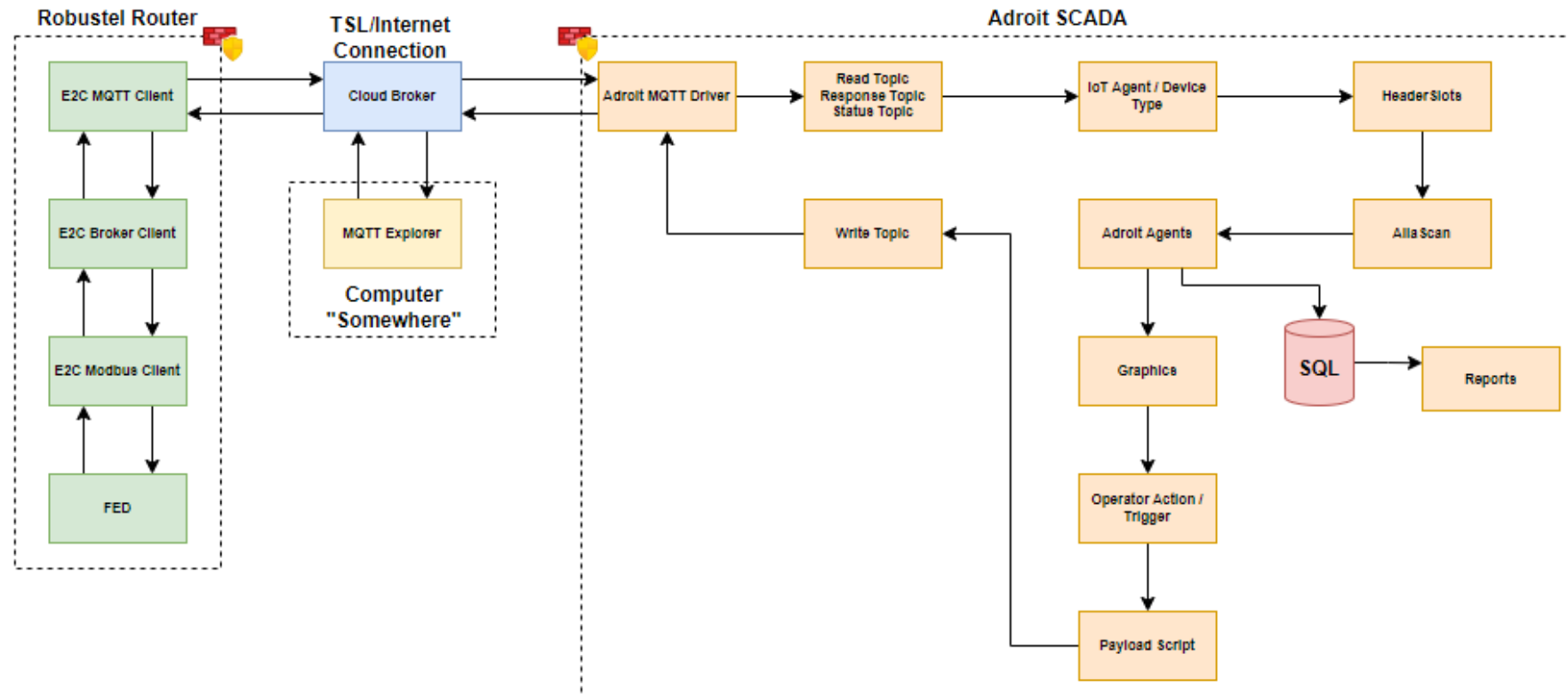
Currently scanned tags: Scan Unscan

Tag >	Address	Rate	Deadband	IO flags
ADR_FX5_F001_AI_D0.rawValue	@ADR_FX5_F001.AIN_D0	0	0	
ADR_FX5_F001_AI_D1.rawValue	@ADR_FX5_F001.AIN_D1	0	0	
ADR_FX5_F001_AI_D2.rawValue	@ADR_FX5_F001.AIN_D2	0	0	
ADR_FX5_F001_AI_D3.rawValue	@ADR_FX5_F001.AIN_D3	0	0	
ADR_FX5_F001_AO_D10.rawValue...	@ADR_FX5_F001.AOT_D10	0	0	
ADR_FX5_F001_AO_D11.rawValue...	@ADR_FX5_F001.AOT_D11	0	0	
ADR_FX5_F001_AO_D12.rawValue...	@ADR_FX5_F001.AOT_D12	0	0	
ADR_FX5_F001_AO_D13.rawValue...	@ADR_FX5_F001.AOT_D13	0	0	
ADR_FX5_F001_DEVICE_INFO....	@ADR_FX5_F001.onlinetag	0	0	
ADR_FX5_F001_DEVICE_INFO....	@ADR_FX5_F001.agentDescrip...	0	0	
ADR_FX5_F001_DEVICE_INFO....	@ADR_FX5_F001.payload1dev...	0	0	
ADR_FX5_F001_DEVICE_INFO....	@ADR_FX5_F001.payload1sub...	0	0	
ADR_FX5_F001_DI_STS.rawValue...	@ADR_FX5_F001.DIN_X0_17	0	0	
ADR_FX5_F001_DO_CTRL.rawValue...	@ADR_FX5_F001.DOT_Y0_17	0	0	
ADR_FX5_F001_DT_FORMAT.c...	@ADR_FX5_F001.time	0	0	
MIDM_CP_TRU000_COMS_FAIL...	@MIDM_CP_TRU000.COMS_...	0	0	
MIDM_CP_TRU000_DEVICE_IN...	@MIDM_CP_TRU000.onlinetag	0	0	
MIDM_CP_TRU000_DEVICE_IN...	@MIDM_CP_TRU000.payload1...	0	0	
MIDM_CP_TRU000_DEVICE_IN...	@MIDM_CP_TRU000.payload1...	0	0	
MIDM_CP_TRU000_DT_FORMA...	@MIDM_CP_TRU000.timestamp	0	0	
MIDM_CP_TRU000_OTP_CUR.r...	@MIDM_CP_TRU000.CURRENT	0	0	
MIDM_CP_TRU000_OTP_VOLT....	@MIDM_CP_TRU000.VOLTAGE	0	0	

5. Network Architecture



6. MQTT Communication Architecture



7. Troubleshooting



- RUN, MDM and Signal LEDs should be on.
- The Signal LED should be as stable as possible.



- GSM antenna connected must be connected to the Main SMA connector.
- If an external antenna or signal booster is required, it will be connected to the AUX SMA connector.

Router not registered on the network	Ensure that the correct sim card is installed to sim slot number 1. (Refer to the device management register.xlsx) Ensure that Vodacom network service is currently available and that reception is good the building or kiosk where the router is installed. GSM antenna connected to the Main connector. Antenna placement is important. If possible, move antenna outside of the kiosk or as high as possible. Monitor the router signal indicator. The LED must remain rather solid. Check sim card status and data balance on the RCMS platform and with RJ Connect. Reboot the router and wait on the RUN, MDM & WLAN LEDs to come on. The Router must be registered to the RCMS platform, initialised and a RCMS licence allocated to it from accounts.
Router not registered on the VPN Network	RCMS service not started on the router. Use remote access from RCMS platform to enable it. The router must be grouped with the other related routers to the same VPN group. Initialise the VPN connection from the RCMS platform and refresh the page. The Router must be registered to the RCMS platform, initialised and a RCMS licence allocated to router serial number from accounts.
Router Remote Access	Device must be online Delete old link and generate a new one.

Router/Client Can't Connect with the Broker	Verify the Broker login credentials, that TLS is enabled, that broker certificate is still valid and applied to the client. Check internet connectivity. Verify if other devices or clients also experience a connection problem, or do other clients still connect with the broker successfully. Login to the broker and see that broker subscription & service is still active and that we have not reached max client connections or that the device sim ran out of data.
Modbus Register/s not updating	Verify that Modbus Client is enabled. Verify Modbus table, IP address, Port, TCP selection, Modbus function and registers to be configured correctly. Check PLC connection and availability. Use Router ping tools to poll the local devices connected to its LAN0 port. Verify that the PLC's Modbus routing table is configured and setup correctly.

SITE COMMISSIONING SHEET

SCADA Engineer	
DATE	
SITE NAME	
REGION	
MQTT REPORT RATE	
MODBUS READ RATE	
FIREWALL STATUS	

TAG	AGENT	RAW & OFF STATE	ENG & ON STATE	DISPLAY CHECKED	UNIT	ALARM ROUTING	TRENDING	DB LOGGING
DEVICE SERIAL	FRAME				Text			
SITE NAME	FRAME				Text			
SITE COORDINATES	FRAME				Text			
DEVICE TIMESTAMP	DATE				Text			
AC MAINS FAIL	MARSHAL/R	AC Mains Fail	AC Mains Healthy		0-1	Critical Alarm/AMA	YES	DBLOG DIGITALS
PSU MAINS FAIL	MARSHAL/R	PSU Mains Fail	PSU Mains Healthy		0-1	Critical Alarm/AMA	YES	DBLOG DIGITALS
DOOR TAMPER	MARSHAL/R	Door Open	Door Close		0-1	Critical Alarm/AMA	YES	DBLOG DIGITALS
CP CONTROL MODE STS	MARSHAL/R	In Auto	In Manual		0-1	Event/AMA	YES	DBLOG DIGITALS
CIRCUIT DISCONNECT RELAY STS	MARSHAL/R	Circuit Closed	Circuit Open		0-1	Warning Alarm/AMA	YES	DBLOG DIGITALS
COMS WARNING	DIGITAL	Coms Healthy	Coms Warning		0-1	Warning Alarm/AMA	YES	DBLOG DIGITALS
COMS FAIL	DIGITAL	Coms Healthy	Coms Fail		0-1	Critical Alarm/AMA	YES	DBLOG DIGITALS
IN/OUT OF SERVICE	DIGITAL	In Service	Out of Service		0-1	Warning Alarm/AMA	YES	DBLOG DIGITALS
CONTROL MODE CTRL	MARSHAL/W	Auto Select	Manual Select		0-1	Event/OprAudit/Security/AMA	NO	DBLOG DIGITALS
CIRCUIT DISCONNECT RELAY CTRL	MARSHAL/W	Circuit Close Select	Circuit Open Select		0-1	Event/OprAudit/Security/AMA	NO	DBLOG DIGITALS
VOLTAGE	ANALOG	0-1000	0-10		V	LL Critical, L Warning/AMA	YES	DBLOG ANALOGS
CURRENT	ANALOG	0-1000	0-100		mA	LL Critical, L Warning/AMA	YES	DBLOG ANALOGS
REFERENCE POTENTIAL	ANALOG	-100-0	-10-0		V	LL Critical, L Warning/AMA	YES	DBLOG ANALOGS
REFERENCE POTENTIAL SP	ANALOG	0-16000	-10-0		V	Event/OprAudit/Security/AMA	YES	DBLOG ANALOGS
PAYLOAD SCRIPT & TRIGGERS	TEST WRITES							
COMS FAIL SCRIPT	TEST COMS FAIL							
AMA REPORT	TEST AMA REPORT							

Hereby we declare that the site has been tested end-to-end—from the frontend device through to the SCADA graphics—and is accepted as fully commissioned

ADROIT SIGNATURE		DATE	
CUSTOMER SIGNATURE		DATE	

Outstanding items

- Adroit: SCADA Base Router & configuration still to be deployed on uMhgeni uThukela Midmar SCADA. To be planned & scheduled with IT during a planned visit.